[INCH-POUND]
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FEDERAL SPECIFICATION

BITS, WOOD-BORING; AND CHISELS, MORTISING, HOLLOW

The General Services Administration has authorized the use of this Federal Specification by all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hardened steel wood-boring bits and hollow mortising chisels.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, and styles. The wood-boring bits shall be of the following types and classes, as specified (see 6.2):

Type I - Bits
- Class A - Auger, double spur
- Class B - Bell-hangers'
- Class C - Car
- Class D - Expansive, plain
- Class E - Expansive, screw or gear adjusting
- Class F - Forstner, hand
- Class G - Forstner, machine
- Class I - Machine, with machine shank
- Class L - Plug, machine
- Class M - Ship auger, with machine shank, with flat
- Class N - Ship auger, with straight square shank
- Class O - Ship auger, with ring
- Class Q - Bits, center, flat head machine shank

Type III - Auger bit sets
- Class A - In canvas, fire-resistant plastic, or imitation leather rolls
- Class B - In wood, plastic, or metal boxes

Type IV - Bits, center (flat head) in sets
Machine bits in canvas, fire-resistant plastic, or imitation-leather rolls

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Tools and Appliances Commodity Center (6FETE-CO), Washington, DC 20406.

FSC 5133

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2. APPLICABLE DOCUMENTS

2.1 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI)/American Society of Quality Control (ASQC):

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Application for copies should be addressed to the American Society for Quality Control, P.O. Box 3005, 611 E. Wisconsin Avenue, Milwaukee, WI 53201-4606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

3. REQUIREMENTS

3.1 Illustrations. Illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of bits and chisels otherwise conforming to this specification.

3.2 Material. The tool steel used in the manufacture of the bits and chisels shall be of such composition and quality as will provide the highest productive performance.

3.3 Cutting edge. All cutting edges of drills, bits, and chisels shall be machined, sharpened, hardened, and tempered as indicated herein to provide clean and easy boring.

3.3.1 Shank. All bit and chisel shanks shall conform to the shape, sizes, and dimensions specified herein. All tapers on square shank bits shall be symmetrical and shall conform to the dimensions specified in Table I.

3.4 Hardness. Except as otherwise specified herein, all bits and chisels shall show a hardness of 40 to 53 on the Rockwell C scale. Type IV bits shall show a hardness of 47 to 62 on the Rockwell C scale, when tested as specified in 4.4.1.

3.5 Size accuracy.

3.5.1 Bits, other than classes D and E. The diameter of the holes bored by bits other than classes D and E shall not vary more than 1/64 inch from the bit size specified hereinafter when tested as specified in section 4.

3.5.2 Classes D and E bits. The difference in distances measured between the screw point and spur before and after testing as specified in section 4 shall not exceed $\pm$ 1/32 inch for plain expansive bits and 1/64 inch for screw or gear adjusting expansive bits.

3.6 Load test requirements. Bits over 3/8 inch in diameter shall show no permanent deflection when tested as specified in section 4.

3.7 Longitudinal relief. The diameter of all straight bits shall be largest at the cutting head than at the shank end, known as back taper. The back taper shall be adequate to assure satisfactory boring performance when tested as specified in section 4.

3.8 Bow. Bits shall be free from any discernible bow when tested as specified in section 4.

3.9 Hand of operation. Unless otherwise specified (see 6.2), bits shall be for right-hand operation.

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3.10 Finish. Chisel surfaces and the groove or flight and body of bits shall have a polished finish, free from scratches, when examined under bright illumination without magnification.

3.11 Marking. Each item shall be marked with the size, manufacturer's name or identifying symbol, and the state or country of manufacture. The marking shall be engraved, etched, or stamped in such a manner that it will be permanent to the extent that it will remain clear and legible throughout the life of the item.

3.12 Type I bits.

3.12.1 Class A, bits, auger.

3.12.1.1 Construction. Unless a specific twist style is specified (see 6.2), class A bits shall be solid center single twist (see Figure 1), double twist (see Figure 2), or single twist (see Figure 3), at the option of the contractor. Solid center single twist and double twist bits shall have a shank, screw point, one or two spurs, and one or two extension lips so beveled as to form cutting edges. Single bits shall have a shank, screw point, extension or side lip, and a spur. Bits shall be either forged from a single piece of steel or securely welded at shank end.



Figure 1. Type I, class A, auger bits and type I, class C car bits.



Figure 2. Type I, class A, auger bits and type I, class C, car bits.



Figure 3. Type I, class A, auger bits and type I, class C, car bits.

3.12.1.2 Screw point. Unless a specific screw point is specified (see 6.2), the screw point shall be either single or double thread and either fine or medium thread (see 6.1).

3.12.1.3 Cutting edges. The cutting edges shall be located at proper distances below the spurs to ensure the best performance of the bit head.

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3.12.1.4 Shank. Class A auger bits shall have a square tapered bit stock shank and comply with the shank dimensions of Table I.

3.12.1.5 Sizes and dimensions. Class A auger bits shall conform to the dimensional requirements of Table I for the sizes specified (see 6.2).

TABLE I. Type I, class A, auger bits

Size (bit diameter) (Inches)	Length		Spur height (Inch)	Diameter of shank round minimum (Inch)	Dimensions of square shank		
	Overall (Inches)	Twist Minimum (Inches)			Large end (Inch)	Small end (Inch)	Length (Inches)
1/4	7-1/2	3-3/8	5/64	13/64	5/16	3/16	1-3/16
5/16	7-3/4	3-1/2	7/64	7/32	5/16	3/16	1-3/16
3/8	7-3/4	3-1/2	1/8	15/64	11/32	3/16	1-1/4
7/16	7-7/8	3-1/2	1/8	17/64	11/32	3/16	1-1/4
1/2	8	3-1/2	9/64	17/64	11/32	3/16	1-1/4
9/16	8-1/8	3-1/2	5/32	9/32	3/8	7/32	1-1/4
5/8	8-1/4	3-1/2	5/32	19/64	3/8	7/32	1-9/32
11/16	8-3/8	4	3/16	19/64	3/8	7/32	1-9/32
3/4	8-1/2	4	3/16	5/16	3/8	7/32	1-5/16
13/16	8-5/8	4-1/4	13/64	5/16	13/32	7/32	1-3/8
7/8	8-3/4	4-1/4	3/16	5/16	13/32	7/32	1-3/8
15/16	8-7/8	4-1/4	3/16	11/32	13/32	7/32	1-3/8
1	9	4-1/2	7/32	11/32	13/32	7/32	1-3/8
1-1/16	9-1/8	4-3/4	17/64	3/8	7/16	1/4	1-3/8
1-1/8	9-1/4	4-3/4	9/32	3/8	7/16	1/4	1-3/8
1-3/16	9-1/4	4-7/8	9/32	3/8	7/16	1/4	1-3/8
1-1/4	9-1/4	4-7/8	9/32	27/64	7/16	1/4	1-3/8
1-5/16	9-3/8	4-7/8	5/16	27/64	7/16	1/4	1-3/8
1-3/8	9-3/8	5	5/16	27/64	7/16	1/4	1-3/8
1-7/16	9-3/8	5	5/16	27/64	7/16	1/4	1-3/8
1-1/2	9-3/8	5	5/16	27/64	7/16	1/4	1-1/2

Tolerances:
Bit Diameter: + 0.015 inch, - 0.000 inch
Length Overall: ± 10 percent of the respective lengths specified
Height of spur: ± 1/32 inch
Square dimension of large and small ends: ± 1/32 inch
Length of square: ± 3/16 inch

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3.12.1.6 Spur measurement. Spur heights for type I bits shall conform to the measurements of Table I when measured from the base, which is on the angle back of the cutting edge, straight up along the center line to the highest point of the crest of the spur (see Figure 4).

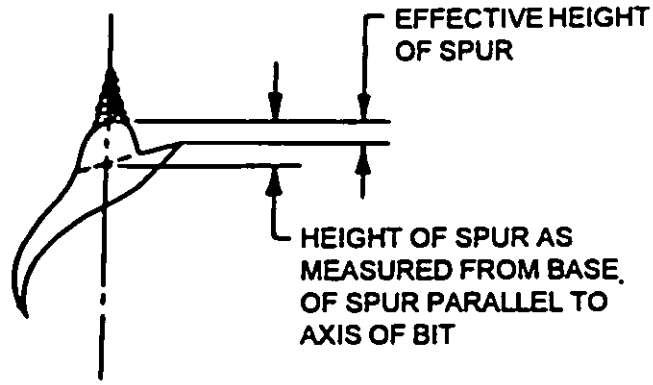


Figure 4. Type I, spur measurement.

3.12.1.7 Twist measurement. Twist or flute lengths shall conform to the requirements of Table I when measured from the cutting edge of the bit to the extreme back end of the twist or flute, or where the twist terminates on the shank diameter.

3.12.2 Class B, bits, bell-hangers'.

3.12.2.1 Construction. Class B bits shall have a double twist, the length of which shall be not less than 3-1/2 inches and the end or head pointed and ground to an included angle of 82° ± 5°. Bits shall be either forged from a single piece of steel or securely welded at shank end. Hole in twist for pulling wire shall be at manufacturer's option.

3.12.2.2 Sizes and dimensions. Bell-hangers' bits shall be similar to Figure 5. Bits shall be furnished in sizes from 3/16 to 3/4 inch, inclusive, by sixteenths of an inch, and in overall lengths of 12, 18, or 24 inches, as specified (see 6.2). Dimensions shall be within the applicable tolerances specified in the footnote of Table I.



Figure 5. Type I, class B, bell-hangers' bits.

3.12.3 Class C, bits, car. Class C bits shall conform to type I, class A, except that those above 3/8 inch diameter shall withstand the test specified in section 4. Class C bits shall be furnished in sizes from 1/4 to 1-1/4 inches, inclusive, in increments of 1/16 inch, with a minimum overall length of 15-3/4 inches and a length of twist of 12 inches minimum for all sizes. Bits shall be similar to Figures 1, 2, or 3 (except the twisted portion is relatively longer than shown).

3.12.4 Class D, bits, expansive, plain.

3.12.4.1 Construction. Class D bits shall consist of a body formed with the shank, a clamping device, clamping screw, and two interchangeable and adjustable cutters suitable for boring holes of any diameter within the capacity range of the bit.

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3.12.4.1.1 Shank. Class D bits shall have a square tapered bit-stock shank and comply with the shank dimensions specified in Table II.

3.12.4.2 Body. The body of the expansive bit shall have a threaded screw point, a single cutting edge, and be either with or without a spur. The throat of the bit shall be open to prevent chips from choking in the throat. An index mark shall be stamped on the body or locking plate of the expansive bit as a regulating point.

3.12.4.3 Clamping device. The cutters or blades shall be held securely by means of a wedging action of a clamping device resulting from tightening the clamping screw. The device shall have sufficient rigidity so as to hold the cutter or blade firmly in place, without creeping during the process of boring. This design shall be such as to permit rapid and accurate adjustment of the cutters or blade so as to lock in the position required.

3.12.4.4 Cutters or blades. Cutters or blades shall fit the body of the bit and shall be provided with a single spur for engaging the work and a ground lip. The cutting edge shall be located below the spur. The cutters or blades shall be permanently marked with equally spaced legible graduations to permit rapid and accurate adjustment. The graduations shall be in increments of 1/32 inch.

3.12.4.5 Sizes and dimensions. Class D, plain expansive bits shall be similar to Figure 6 and conform to dimensions shown in Table II for sizes specified (see 6.2).

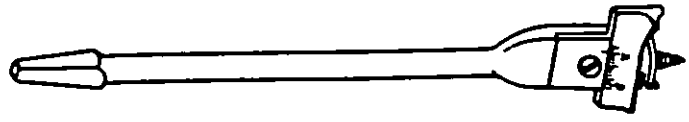


Figure 6. Type I, class D, plain expansive bits.

TABLE II. Type I, class D, plain expansive bits

Size (expansive) range Minimum (Inches)	Overall length		Cutter spur height (Inch)	Diameter of shank round Minimum (Inch)	Dimensions of square shank		
	Minimum (Inches)	Maximum (Inches)			Large end (Inch)	Small end (Inch)	Length (Inches)
1/2 to 1-1/2	7	9	11/64	5/16	3/8	7/32	1-1/4
7/8 to 3	8-1/2	10-1/2	3/16	3/8	7/16	1/4	1-3/8

Tolerances:
Height of spur: ± 1/64 inch
Square shank dimensions of large and small ends: ± 1/32 inch
Length of square shank: ± 1/4 inch

3.12.5 Class E, bits, expansive, screw or gear adjusting.

3.12.5.1 Construction. Bits shall have a body formed integral with the shank, a clamping device, adjusting gear or screw, clamping screw; and by forging or by welding at least two cutters suitable for boring holes of any diameter within the capacity range of the bit.

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3.12.5.2 Body. The body of the expansive bit shall have a threaded screw point, a single cutting edge, and be either with or without a spur.

3.12.5.3 Clamping device. The cutters or blades shall be held securely to the body of the expansive bit by means of a screw or gear which shall mesh with the teeth of the cutter or blade. This device shall have sufficient rigidity so as to hold the cutter or blade firmly in place without creeping during the process of boring. The design shall be such as to permit rapid and accurate adjustment of the cutters or blades.

3.12.5.4 Cutters or blades. The cutters or blades shall fit accurately into the body of the bit and shall be provided with a single spur for engaging in the work and a ground lip. The cutting edge shall be in advance of the spur. The cutters or blades shall have cut teeth or threads forming a rack into which the adjusting device shall mesh.

3.12.5.5 Marking for diameter of adjustment. The expansive bit or cutter shall be permanently marked with equally spaced legible graduations indicating the diameter of the adjustment. The graduations shall be in increments of 1/32 inch.

3.12.5.6 Sizes and dimensions. Class E screw or gear adjusting expansive bits shall be similar to Figure 7 and conform to the dimensions shown in Table III for sizes specified (see 6.2).

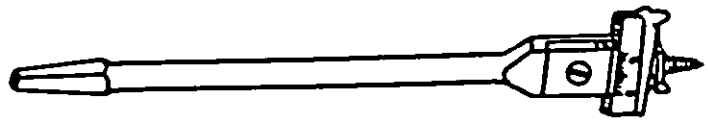


Figure 7. Type I, class E, screw or gear adjusting expansive bits.

TABLE III. Type I, class E, bits, expansive, screw or gear adjusting

Size (expansive) range Minimum (Inches)	Overall length		Cutter spur height (Inch)	Diameter of shank round			
	Minimum (Inches)	Maximum (Inches)		Minimum (Inch)	Large end (Inch)	Small end (Inch)	Length (Inches)
5/8 to 1-3/4	7-1/2	9	11/64	5/16	3/8	7/32	1-1/4
7/8 to 3	8-1/2	10-1/2	3/16	3/8	7/16	1/4	1-3/8

Tolerances:
Height of spur: ± 1/32 inch
Square shank dimensions of large and small ends: ± 1/32 inch
Length of square shank: ± 1/4 inch

3.12.6 Class F, bits, forstner, hand.

3.12.6.1 Construction. Class F bits shall be guided by a circular rim in lieu of a screw point. Slots in the rim shall be located in advance of the cutting edge and shall be of ample size to allow free egress of chips without heating bit excessively. The cavity within the head shall be smooth and free from all obstruction to free egress of chips. Round shanks of bits 1/4 and 5/16 inch in diameter shall have a spiral flute for a distance of approximately 2-1/4 inches long to provide sufficient lip clearance.

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3.12.6.2 Cutting lips. Bits 1/4 to 7/16 inch in diameter shall have a single cutting lip, and sizes 1/2 inch and larger shall have two cutting lips.

3.12.6.3 Shank. Class F bits shall have square tapered bit stock shanks 1-7/16 inches long by 7/16 inch square at the large end with a taper of 4-1/2° on a side.

3.12.6.4 Sizes and dimensions. Class F bits shall be similar to Figure 8 and comply with the dimensions shown in Table IV for sizes specified (see 6.2).

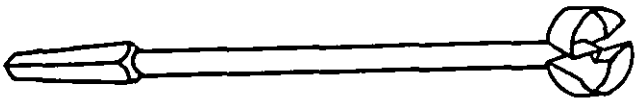


Figure 8. Type I, class F, hand forstner bits.

TABLE IV. Type I, class F, bits, forstner, hand

Diameter of cutting head (Inches)	Length overall (Inches)	Diameter of shank round Minimum (Inch)
1/4	6-1/4	7/32
5/16	6-1/4	7/32
3/8	6-1/4	1/4
7/16	6-1/4	1/4
1/2	6-1/4	9/32
9/16	6-3/4	9/32
5/8	6-3/4	9/32
11/16	6-3/4	9/32
3/4	6-3/4	9/32
13/16	7-1/4	5/16
7/8	7-1/4	5/16
15/16	7-1/4	5/16
1	7-3/4	5/16
1-1/16	7-3/4	11/32
1-1/8	7-3/4	11/32
1-3/16	7-3/4	11/32
1-1/4	7-3/4	3/8
1-5/16	8-1/4	3/8

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TABLE IV. Type I, class F, bits, forstner, hand (cont'd)

Diameter of cutting head (Inches)	Length overall (Inches)	Diameter of shank round Minimum (Inch)
1-3/8	8-1/4	3/8
1-7/16	8-1/4	1/2
1-1/2	8-1/4	1/2
1-5/8	8-1/4	1/2
1-3/4	8-3/4	1/2
1-7/8	8-3/4	1/2
2	8-3/4	1/2

Tolerances:
Diameter: + 0.015 inch, - 0.000
Length overall: ± 3/8 inch
Square shank dimensions of large and small ends: ± 1/32 inch
Length of square shank: ± 1/4 inch

3.12.7 Class G, bits, forstner, machine. Class G bits shall conform to type I, class F bits, except as follows:

3.12.7.1 Shank. Bits shall have round, straight shanks 1/2 inch in diameter by a minimum of 2 inches long. Unless otherwise specified (see 6.2), the shank shall be flattened on one side to a width of 3/8 inch and length of 1-1/2 inches.

3.12.7.2 Overall length. The overall length for diameters 1/4 to 1-3/8 inches, inclusive, shall be 5 ± 1/4 inches, and for diameters 1-7/16 to 2 inches, inclusive, the overall length shall be 6-3/4 ± 1/4 inches.

3.12.7.3 Shape. Machine forstner bits shall be similar to Figure 9.

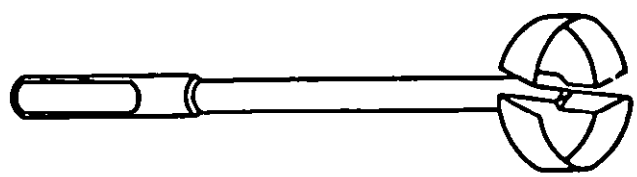


Figure 9. Type I, class G, machine forstner bits.

3.12.8 Class J, bits, machine, with machine shank.

3.12.8.1 Construction. Unless a specific twist is specified in the contract or order (see 6.2), class I bits shall be machined drill twist (see Figure 10), solid center (see Figure 11), or wood-bit twist (see Figure 12), at the option of the contractor. The bits shall have a pyramid-type square point and two spurs. Machine drill twist bits and wood-bit twist bits shall be double twist and shall have two cutting edges; solid center bits (see Figure 10) shall be single twist and the head shall be machined so as to produce two cutting edges, the supporting twist of the additional cutting edge shall terminate directly back of the head.

3.12.8.1.1 Shank. Class I bits shall have round straight shanks 1/2 inch in diameter, 2 inches long, without, with one, or with three flats, as specified (see 6.2).

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Figure 10. Type I, class I, machine bits, with machine shank, machine drill twist.



Figure 11. Type I, class I, machine bits, with machine shank, solid center.



Figure 12. Type I, class I, machine bits, with machine shank, wood bit twist.

3.12.8.2 Sizes and dimensions. Class I bits shall conform to the dimensions shown in Table V for sizes specified (see 6.2).

TABLE V. Type I, class I, machine bits, machine shank, without, with one, or with three flats

Size (bit diameter) (Inches)	Length												Spur height (Inch)
	Twist (Inches)						Overall (Inches)						
1/4	4	6	8				6-1/4	8-1/4	10-1/4				5/64
5/16	4	6	8	10			6-1/4	8-1/4	10-1/4	12-1/4			1/8
3/8	4	6	8	10	12		6-1/4	8-1/4	10-1/4	12-1/4	14-1/4		1/8
7/16	4	6	8	10	12		6-1/4	8-1/4	10-1/4	12-1/4	14-1/4		9/64
1/2	4	6	8	10	12		6-1/4	8-1/4	10-1/4	12-1/4	14-1/4		9/64
9/16	4	6	8	10	12	14	6-1/4	8-1/4	10-1/4	12-1/4	14-1/4	16-1/4	11/64
5/8	4	6	8	10	12		6-1/4	8-1/4	10-1/4	12-1/4	14-1/4		11/64
11/16	4	6	8	10	12	14	6-1/4	8-1/4	10-1/4	12-1/4	14-1/4	16-1/4	3/16
3/4	4	6	8	10	12		6-1/4	8-1/4	10-1/4	12-1/4	14-1/4		3/16
13/16	4	6	8	10	12	14	6-1/4	8-1/4	10-1/4	12-1/4	14-1/4	16-1/4	13/64
7/8	4	6	8	10	12		6-1/4	8-1/4	10-1/4	12-1/4	14-1/4		1/4

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TABLE V. Type I, class I, machine bits, machine shank, without, with one, or with three flats (cont'd)

Size (bit diameter) (Inches)	Length												Spur height (Inch)
	Twist (Inches)						Overall (Inches)						
15/16	4	6	8	10	12	14	6-1/4	8-1/4	10-1/4	12-1/4	14-1/4	16-1/4	1/4
1	4	6	8	10	12		6-1/4	8-1/4	10-1/4	12-1/4	14-1/4		1/4
1-1/16	4	6	8	10	12	14	6-1/4	8-1/4	10-1/4	12-1/4	14-1/4	16-1/4	17/64
1-1/8	4	6	8	10	12		6-1/4	8-1/4	10-1/4	12-1/4	14-1/4		9/32
1-3/16	4	6	8	10	12		6-1/4	8-1/4	10-1/4	12-1/4	14-1/4		9/32
1-1/4	4	6	8	10	12		6-1/4	8-1/4	10-1/4	12-1/4	14-1/4		9/32

Tolerances:
Bit diameter : + 0.010 inch, - 0.000 inch
Length overall: ± 1/4 inch
Height of spur: ± 1/32 inch
Shank diameter: + 0.000 inch, - 0.005 inch

3.12.9 Class L, bits, plug,, machine. Class L, machine plug bits shall have one or two spurs, and one or two extension lips so beveled as to form cutting edges. The bit shall be guided by a pilot (see 3.12.9.2).

3.12.9.1 Shank. Bits shall have round straight shanks 1/2 inch in diameter by 2 inches long. Unless otherwise specified (see 6.2), the shank shall be flattened on one side to a width of 3/8 inch and to a length of 1-1/2 inches.

3.12.9.2 Pilot. The bit shall have cylindrical pilot, solid, or inserted and secured by a set screw axially located in lieu of a screw point.

3.12.9.3 Sizes and dimensions. Class L, machine plug bits shall be similar to Figure 13 and conform to the dimensions shown in Table VI for sizes specified (see 6.2).

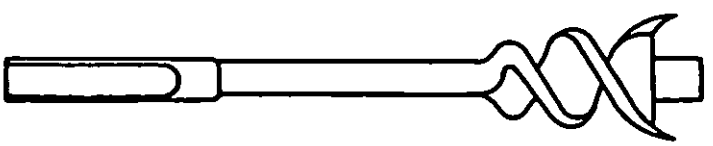


Figure 13. Type I, class L, machine plug bits.

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TABLE VI. Type I, class L, bits, plug, machine

Size (bit diameter) (Inches)	Length		Dimensions of pilot		Spur height (Inch)	Diameter of shank round Minimum (Inch)
	Overall (Inches)	Twist Minimum (Inches)	Diameter (Inch)	Length (Inch)		
1-1/8	6	2	1/2	1/2	9/32	3/8
1-1/8	6	2	9/16	1/2	9/32	3/8
1-1/4	6	2	5/8	1/2	9/32	7/16

Tolerances:
Bit diameter : + 0.010 inch, - 0.000 inch
Pilot diameter: + 0.000 inch, - 0.008 inch
Length overall: ± 1/8 inch
Height of spur: ± 1/64 inch
Guide length: ± 1/4 inch

3.12.10 Class M, bits, ship auger, with machine shank, with flat or hexagon shank for use with 1/2 inch socket, three jaw chuck and hand braces.

3.12.10.1 Construction. Class M bits shall be of the single-twist type. Unless specifically required (see 6.2), at the option of the procuring agency, these bits may be furnished with or without a solid center. Bits without solid centers shall have a cutting edge and side lip located at right angles to the cutting edge. Bits shall have screw points unless otherwise specified.

3.12.10.2 Shank. Bits shall have either round straight shanks 1/2 inch in diameter by 2 inches long or hexagon shank 1/2 inch across the flats. Unless otherwise specified (see 6.2), the round shank shall be flattened on one side to a width of 3/8 inch and to a length of 1-1/2 inches.

3.12.10.3 Sizes and dimensions. Class M, ship auger bits with no point or with screw point shall be similar to Figure 14 and conform to the dimensions shown in Table VII for sizes specified (see 6.2).



Figure 14. Type I, class M, machine shank, ship auger bits.

TABLE VII. Type I, class M, bits, ship auger, with machine shank, with flat

Size (bit diameter) (Inch)	Length		Diameter of shank round Minimum (Inch)
	Overall (Inches)	Twist Minimum (Inches)	
1/4	10-1/4	6	7/32
5/16	12-1/4	8	7/32
3/8	12-1/4	8	1/4
7/16	12-1/4	8	17/64

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TABLE VII. Type I, class M, bits, ship auger, with machine shank, with flat (cont'd)

Size (bit diameter) (Inch)	Length		Diameter of shank round Minimum (Inch)
	Overall (Inches)	Twist Minimum (Inches)	
1/2	12-1/4	8	9/32
9/16	14-1/4	10	19/64
5/8	14-1/4	10	13/32
11/16	14-1/4	10	13/32
3/4	14-1/4	10	13/32
13/16	14-1/4	10	13/32
7/8	14-1/4	10	13/32
15/16	14-1/4	10	13/32
1	14-1/4	10	13/32

Tolerances:
Bit diameter at head: + 0.015 inch, - 0.000 inch
Length overall: ± 1/2 inch

3.12.11 Class N, bits, ship auger, with straight square shank. Class N bits shall be similar to type I, class M, except that the shape of the shank shall be as follows:

3.12.11.1 Shank. Bits shall have either straight square or hexagon shanks of the dimension specified in Table VIII and shall be similar to Figure 14. Shanks shall be suitable for use with auger handles, single or double crank handles, or welding to extension stems.

3.12.11.2 Sizes and dimensions. Class N, ship auger bits with no point or with screw point shall be similar to Figure 14 and conform to the dimensions specified in Table VIII for the sizes specified (see 6.2).

TABLE VIII. Type I, class N, bits, ship auger, with straight square shank or hexagon shank

Size (bit diameter) (Inches)	Length		Type of shank	
	Overall (Inches)	Twist Minimum (Inches)	Square shank Size (Inch)	Hexagon shank Size (Inch)
7/16	14-1/2	10	3/8	5/16
5/8	16-1/4	12	7/16	7/16
9/16	16-1/4	12	7/16	7/16
11/16	16-1/4	12	7/16	7/16
3/4	16-1/4	12	7/16	7/16
13/16	16-1/4	12	7/16	7/16
7/8	18	12	7/16	7/16

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TABLE VIII. Type I, class N, bits, ship auger, with straight square shank or hexagon shank (cont'd)

Size (bit diameter) (Inches)	Length		Type of shank	
	Overall (Inches)	Twist Minimum (Inches)	Square shank Size (Inch)	Hexagon shank Size (Inch)
15/16	18	12	7/16	7/16
1	18	12	7/16	7/16
1-1/16	18	12	5/8	7/16
1-1/8	18	12	5/8	7/16
1-1/4	18	12	5/8	7/16
1-3/8	18	12	5/8	7/16
1-1/2	18	12	5/8	7/16
1-3/4	18	12	5/8	7/16
2	18	12	5/8	7/16

Tolerances:
Bit diameter: + 0.015 inch, - 0.000 inch
Length overall: ± 1 inch
Square shank size: ± 1/32 inch
Hexagon shank diameter: + 0.000 inch, - 0.005 inch

3.12.2 Class O, bits, ship auger, with ring, and with screw point.

3.12.12.1 Construction. Class O, bits shall be solid-center, single-twist, single-twist, or double-twist at the option of the contractor. Bits of the solid-center design shall have two cutting edges and two spurs located at right angles to the cutting edges. Bits of the single-twist design shall have one cutting edge and one side lip at right angles to the cutting edge. Bits of the double-twist design shall have a flat-cut head and two spurs located at right angles to the cutting edge. Bits shall have medium-pitch screw points unless otherwise specified (see 6.2). Bits shall have a round-shank but welded onto a suitable ring, and when specified (see 6.2) in the contract or order, a handle. The sizes of the square shall be at option of the contractor.

3.12.12.2 Shank and ring. The shank shall be fitted with a ring or eye of either 1 inch or 1-1/4 inch inside diameter suitable for reception of a wood handle (see 3.12.12.3) and of such strength as to withstand the test specified in section 4.

3.12.12.3 Handle. The handles shall be of hardwood, two sizes; 1 inch (+ 1/8 inch, - 0 inch), and 1-1/4 inches (+ 1/8 inch, - 0 inch) in diameter by not less than 12 inches ± 1 inch long. Handles for augers to 1 inch in diameter shall have their centers turned with a slight taper so that they can be wedged tightly into the 1 inch diameter hole in the ring, and handles for augers greater than 1 inch in diameter shall have their centers turned with a slight taper so that they can be wedged tightly into the 1-1/4 inch diameter hole in the ring.

3.12.12.4 Sizes and dimensions. Class O bits, with no point or with screw point, shall be similar to Figure 15 and shall conform to the dimensional requirements of Table IX for the sizes specified (see 6.2).



Figure 15. Type I, class O, ring ship auger bits.

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TABLE IX. Type I, class Q, bits, ship auger, with ring

Size (bit diameter) (Inches)	Length		Diameter of shank round Minimum (Inch)
	Approximately overall (Inches)	Twist Minimum (Inches)	
1/2	14-1/2	10	5/16
9/16	17-1/2	12	11/32
5/8	17-1/2	12	11/32
11/16	17-1/2	12	3/8
3/4	17-1/2	12	3/8
13/16	17-1/2	12	3/8
7/8	17-1/2	12	3/8
15/16	17-1/2	12	3/8
1	17-1/2	12	3/8
1-1/16	20	15	15/32
1-1/8	20	15	15/32
1-3/16	20	15	15/32
1-1/4	20	15	15/32
1-5/16	20	15	19/32
1-3/8	20	15	19/32
1-7/16	20	15	19/32
1-1/2	20	15	19/32
1-9/16	20	15	19/32
1-5/8	20	15	19/32
1-11/16	20	15	19/32
1-3/4	20	15	19/32
1-13/16	20	15	19/32
1-7/8	20	15	19/32
1-15/16	20	15	19/32
2	20	15	19/32

Tolerances:
Bit diameter: + 0.015 inch, - 0.000 inch
Length overall: ± 1/2 inch, of standard length

3.12.13 Class Q, bits, center, flat head, machine shank.

3.12.13.1 Construction. Class Q bits shall have a flat head, a center point, and round with three flats, or hexagon shank. The head shall be made of high speed or high carbon steel, and shall have a hardness of 47 to 62 on the Rockwell C scale, and have relieved cutting lips and center points. The cutting sizes shall be furnished in 1/16 inch increments from 1/4 to 1 inch. Sizes 3/8 inch and smaller shall have a shank diameter, a width across the flats of the hex, or a width across the corners of the hex, of 3/16 inch minimum. Sizes over 3/8 inch shall have a shank

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diameter, a width across the flats of the hex, or a width across the corners of the hex, of 1/4 inch minimum. Overall length shall be 6-1/4 inches, ± 1/4 inch.



Figure 16. Class Q, bits, center, flat head, machine shank.

3.13 Type III, auger bit sets.

3.13.1 Class A, auger bit sets, in canvas, fire-resistant plastic or imitation leather rolls. Class A auger bit sets shall consist of one auger bit, type I, class A, or class B of each of the following sizes: 1/4, 3/8, 1/2, 5/8, 3/4, 7/8, and 1 inch. When specified, the set shall consist of 13 type I, class A auger bits in sizes 1/4 through 1 inch in 1/16 inch increments and one type I, class D, plain expansive bit in the 7/8 to 3 inch capacity range. Each set shall be furnished in a flexible roll or case of heavy canvas, fire-resistant plastic, or imitation leather, with a separate pocket for each bit. The roll or case shall be of good quality and workmanship, and provided with a tie tape or buckle so arranged that the set may be easily rolled up and fastened in a secure manner. All bits in any one set shall be similar to Figure 1 or 2.

3.13.2 Class B, auger bit sets, in wood, plastic, or metal boxes. Class B auger bit sets shall consist of one auger bit, type I, class A, of each of the following sizes: 1/4, 5/16, 3/8, 7/16, 1/2, 9/16, 5/8, 11/16, 3/4, 13/16, 7/8, 15/16, and 1 inch. Each set shall be furnished in a substantial wood, plastic, or metal box having a separate hole, groove, or clip for each bit conveniently arranged for easy access. The boxes shall be well constructed, neatly finished, and except for plastic boxes, be coated with shellac or varnish or other substances which preclude corrosion and the absorption of moisture. All bits in any one set shall be similar to Figure 1 or 2.

3.14 Type IV, center bit, flat head, high speed, bit sets.

3.14.1 Type IV sets. Each type IV set shall consist of one each, class Q, of the following sizes: 3/8, 1/2, 5/8, 3/4, 7/8, and 1 inch and shall be similar to Figure 16.

3.14.1.1 Roll or case. Each set of center bits shall be contained in a heavy canvas, fire-resistant plastic, or imitation leather flexible roll or case with a separate pocket for each bit. The roll or case shall be of good quality and workmanship, provided with a tie tape or buckle so arranged that the set may be easily rolled up and fastened in a secure manner.

3.15 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the specification preparing activity for change to this document.

3.16 Workmanship: Details of workmanship shall be in accordance with the best commercial standards and practices. Paints, coatings, platings, and finishes shall be smooth, dry, adherent, continuous, and not stained or discolored. Fasteners shall be firmly secured and show no evidence of deformation, cross threading, or hazardous burrs. Adhesives and lubricants adequate for their intended purpose shall be properly and neatly applied. Adhesives shall be adequately cured. Wires and cables shall be neatly dressed and shall not be frayed or in contact with sharp edges. Wire and cable insulation shall show no evidence of burns, abrading, or pinch marks. There shall be no interference, binding, or galling of parts. External and bearing surfaces shall be free of tool and gouge marks,

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nicks, or other surface imperfections. The item shall be clean and free of corrosion and debris (e.g., chips, shavings, slivers) or other foreign material. The item shall be free from manufacturing workmanship defects (e.g., loose, missing, binding or misaligned parts, sharp or rough external edges, corners, or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specification and standards to the extent specified, or, if none, in accordance with this specification.

4.2 Quality conformance inspection. A random sample of bits shall be selected from each lot in accordance with ANSI/ASQC Z1.4. Data for sampling shall be as specified in Table X.

TABLE X. Sampling

Category	Inspection level	Acceptable quality level (AQL) (percent defective)
Visual, dimensional, and workmanship	S-4	4.0
Hardness and other physical testing	S-3	4.0

4.2.1 Examination. Each of the sample bits or chisels selected in accordance with 4.2 shall be visually and dimensionally examined to verify compliance with this specification. Any bit or chisel in the sample containing one or more defects shall not be offered for delivery. If the number of defective bits or chisels in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.2.2 Bow. Each sample bit (except type IV flat head) shall be tested for bow. Roll bit on hard flat surface between light and eye. If any bow or runout is discernible between the surface and bit, the sample bit does not comply with this specification and shall be rejected.

4.3 Quality conformance tests. Each sample bit or chisel selected in accordance with 4.2.2 shall be subjected to the tests of 4.4.1 through 4.4.4. Any bit or chisels continuing one or more defects shall not be offered for delivery. If the number of defective bits or chisels in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

4.4 Tests.

4.4.1 Hardness. Sample bits or chisels shall be tested to determine compliance with the hardness requirements specified in 3.4. The hardness shall be determined on a Rockwell tester, or an equivalent tester employing a diamond penetrator, and a 150 kilogram major load used on opposite edges of the bit, at or adjacent to the cutter head.

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4.4.2 Accuracy of size.

4.4.2.1 Bits other than classes D and E expansive bits. A hole shall be bored through not less than 4 inches of oak, ash, or maple hardwood, and the diameter shall be measured at both ends to determine conformance with requirements listed herein.

4.4.2.2 Classes D and E expansive bits. A hole shall be bored through not less than 4 inches of oak, ash, or maple hardwood with each cutter at both the maximum and minimum extension to verify conformance with requirements listed herein.

4.4.3 Load tests. The sample bit does not comply with this specification if after applying the applicable load test requirements of 4.4.3.1 a permanent deflection results.

4.4.3.1 Auger I, classes A, I, and gimlet bits. The sample bits of type I, classes A and I shall be clamped to a depth of 2 inches (head end or shank end) in a vise or suitable cantilever type of support and the load specified in Table XI gradually applied at a point not more than 1 inch from the opposite end in a direction perpendicular to its longitudinal axis. The sample bits of type I, classes A and I shall be clamped to a depth of 2 inches (head end or shank end) in a vise or suitable cantilever type of support and the load specified in Table XI gradually applied at a point not more than 1 inch from the opposite end in a direction perpendicular to its longitudinal axis. The load for type I, classes B, C, and O shall be gradually applied at a point 6 ± 1/4 inches from the face of the support device.

TABLE XI. Load tests

Size (bit diameter) (Inches)	Load (Pounds)
7/16 to 9/16	40
5/8 to 3/4	50
13/16 to 15/16	75
1 to 1-1/8	100
1-3/16 to 1-5/16	125
1-3/8 to 1-1/2	150

4.4.4 Ship auger ring test. The shank of the bit shall be clamped firmly in a vise or suitable cantilever type of support and a handle 12 inches long inserted midway into the ring in a horizontal position. A load of 150 pounds shall be gradually applied at a point 1 ± 1/8 inch from the end of the handle. The sample bit does not comply with this specification if after applying the above test the ring is distorted or loosened from the shank.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with Table XII. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

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TABLE XII. Classification of preparation for delivery defects

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling
	Distortion of container
Contents	Number per container is more or less than required
	Net weight exceeds requirements

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be as specified in the contract or purchase order.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The bits and mortising chisels described in this specification are intended for use as wood-working tools for drilling holes and mortising operations. Auger bits may be obtained with the following screw points: fine, for cabinet work; medium, for general purpose boring and ordinary work; and coarse, for rapid operations, particularly in rough work. Table XIII indicates the screw points ordinarily available.

TABLE XIII. Available screw points

Bit diameter (Inch)	Approx. number of threads (per inch)	Type of thread	Approx. number of thread (on point)	Bit diameter (Inch)	Approx. number of threads (per inch)	Type of thread	Approx. number of thread (on point)
			Fine Thread				
3/16	32	Double	4	5/8	24	Double	7
1/4	30, 32	Double	6	11/16	24	Double	8
5/16	28	Double	6	3/4	24	Double	8
3/8	24	Double	6	13/16	24	Double	9
7/16	24	Double	6	7/8	24	Double	10
1/2	24	Double	6	15/16	24	Double	10
9/16	24	Double	7	1	24	Double	10

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TABLE XIII. Available screw points (cont'd)

Bit diameter (Inch)	Approx. number of threads (per inch)	Type of thread	Approx. number of thread (on point)	Bit diameter (Inch)	Approx. number of threads (per inch)	Type of thread	Approx. number of thread (on point)
			Medium Thread				
3/16	32	Double	4	9/16	14, 15, 16	Single	5
1/4	28, 30, 32	Double	6	5/8	14, 15, 16	Single	5
1/4	18	Single	6	11/16	14, 15, 16	Single	5
5/16	24, 26, 28	Double	6	3/4	14, 15, 16	Single	6
5/16	18	Single	6	13/16	14, 15, 16	Single	6
3/8	28	Double	6	7/8	16	Single	6
3/8	12	Single	6	15/16	16	Single	6
7/16	28	Double	6	1	16	Single	6
7/16	14, 16	Single	6				
1/2	20	Double	6				
1/2	14	Single	6				
			Coarse Thread				
3/16	20	Single	3	5/8	10 to 14	Single	3
1/4	18	Single	3	11/16	10 to 14	Single	3
5/16	16	Single	3	3/4	10 to 14	Single	3
3/8	14	Single	3	13/16	10 to 14	Single	4
7/16	12	Single	3	7/8	10 to 14	Single	4
1/2	12, 13, 14	Single	3	15/16	10 to 14	Single	4
9/16	12, 13, 14	Single	3	1	10 to 14	Single	4

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type, class, and style, if applicable, of item required (see 1.2).
c. Marking, if other than specified.
d. If other than right-hand bit is required (see 3.9)
e. For type I.
   (1) If other than flattened shank is required on classes G, I, L, and M bits, (see 3.12.7.1, 3.12.8.1.1, 3.12.9.1, and 3.12.10.2).
   (2) When handle is required for class O (see 3.12.12.1).
   (3) In addition to the above, purchasing officers may exercise the options contained in 3.12.1.1, 3.12.1.2, 3.12.5.2, 3.12.8.1, 3.12.10.1, 3.12.12.1, and 3.12.12.4.
f. The name of the manufacturer and style number of machine on which the chisels are to be used, and the dimensions of the chisel.
g. Level of preservation, packaging, and packing required (see 5.1).

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MILITARY INTERESTS:

Custodians
Army - GL
Air Force - 99

User Activity
Army - CE, ME
Marine - MC
Navy - YD

Preparing Activity:

GSA - FSS

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